Danh mục: medication

Type 2 diabetes in adults: management

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NICE guideline
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Recommendations for research
The guideline committee has made the following recommendations for research.
Key recommendations for research
1 Treatment strategy for people with type 2 diabetes and frailty
For people with type 2 diabetes and frailty, what is the clinical and cost effectiveness of different treatment strategies compared with usual care?
[2026]
Why the committee made the recommendation for research
Because of concerns about adverse effects and polypharmacy, the committee agreed that SGLT-2 inhibitors may not be appropriate for some people with frailty and type 2 diabetes.
There was no specific evidence for people with frailty, so the committee could not recommend a particular method of assessment or cutoff for prescribing SGLT-2 inhibitors. The decision would need to be made based on clinical judgement, taking into account the needs of each person.
The committee recommended medicines for this group based on:
their own expertise
common medicine contraindications in this group
their knowledge of which medicines were likely to have the most manageable side effects.
The committee did not recommend GLP-1 receptor agonists and tirzepatide for people with frailty. However, they agreed that there is no additional safety risk for this population. Therefore, if a person has a relevant indication and frailty, they can still be offered a GLP-1 receptor agonists or tirzepatide.
Full details of the evidence and the committee's discussion are in:
evidence review E: initial management
evidence review F: subsequent management
2 Access to SGLT-2 inhibitors
How can prescribing and access to SGLT-2 inhibitors be improved for people with type 2 diabetes from the most deprived groups?
What factors influence healthcare professionals' decisions about prescribing SGLT-2 inhibitors to adults with and without early onset type 2 diabetes?
What are the most effective and cost-effective methods to increase access and uptake of SGLT-2 inhibitors for people with and without early onset type 2 diabetes who are underserved in the current service?

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[2026]

Why the committee made the recommendation for research

The committee reviewed real-world evidence that SGLT-2 inhibitors are under-prescribed, particularly to women and older people, people from some ethnic backgrounds, and people who have experienced higher levels of deprivation when sex and age are accounted for. They agreed that further research is needed to understand the reasons behind this so made a recommendation for research on improving access to SGLT-2 inhibitors.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

3 Treatments for people with early onset type 2 diabetes

What is the clinical and cost effectiveness of GLP-1 receptor agonists or tirzepatide with SGLT-2 inhibitors compared to SGLT-2 inhibitors alone and to placebo alone for people with early onset type 2 diabetes who are taking metformin?

[2026]

Why the committee made the recommendation for research

The committee recommended combining a GLP-1 agonist or tirzepatide with metformin and an SGLT-2 inhibitor based on evidence from a pooled network meta-analysis and their clinical experience.

The evidence in the pooled network meta-analysis used in the health economic modelling came from a review of people at higher risk of developing cardiovascular disease or people with existing atherosclerotic cardiovascular disease adding subsequent therapies to previous treatment. It showed benefits from GLP-1 receptor agonists, but most studies did not give separate results based on the number or type of other treatments received. A small number of studies included triple therapy combining GLP-1 receptor agonists, SGLT-2 inhibitors and metformin. When compared in health economic evaluation, adding most GLP-1 receptor agonists was not cost effective, while adding liraglutide to an SGLT-2 inhibitor and metformin reported an incremental cost-effectiveness ratio (ICER) approaching £20,000 per quality-adjusted life year (QALY) gained. Tirzepatide was not analysed for this population.

The evidence for combination therapy with metformin and SGLT-2 inhibitors showed that the cardiovascular benefits came from the SGLT-2 inhibitors alone. This was clear because the people receiving metformin and placebo did not get the same benefits. When compared with placebo in clinical trials, GLP-1 receptor agonists also showed cardiovascular benefits regardless of other treatment received. The evidence evaluated for tirzepatide did not show cardiovascular benefits, which leaves some uncertainty about its use for this purpose. However, the committee acknowledged the benefits in reducing HbA1c and weight, and how this could lead to beneficial cardiovascular outcomes in the long term.

The committee agreed that GLP-1 receptor agonists and tirzepatide should be considered in addition to metformin and SGLT-2 inhibitors, given the:

relatively small size of this group

health inequalities that this group would face if they did not receive treatment early, and

challenges in identifying appropriate data.

The committee also made a recommendation for research on treatments for people with early onset diabetes.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

4 The effects of stopping or switching medicines to control blood glucose levels

In adults with type 2 diabetes, what are the effects of stopping and/or switching medicines to control blood glucose levels, and what criteria should inform the decision?

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[2015]

Why the committee made the recommendation for research

Why this is important

There is a lack of evidence on the effects of stopping and/or switching medicines to control blood glucose levels. The current practice of 'stopping rules' is typically motivated by either inadequate blood glucose control (rising HbA1c levels) or intolerable side effects. There is limited understanding of the short- and longterm effects of stopping a therapy and switching to another in terms of diabetes control (HbA1c levels), hypoglycaemic risk, weight gain, and cardiovascular morbidity and mortality. In addition, there is limited understanding of how quickly consideration should be given to stopping and switching to another medicine and, if stopping and switching may be needed, what the optimal sequencing is of medicines. Randomised controlled trials examining these different issues would help to improve diabetes care.

5 Self-monitoring of blood glucose levels

What is the optimal frequency for selfmonitoring of blood glucose in adults with type 2 diabetes?

[2015]

Why the committee made the recommendation for research

Why this is important

There is limited evidence in relation to the long-term effects (at least 5 years) of blood glucose lowering therapies, particularly newer agents in terms of efficacy and adverse events (for example, cardiovascular outcomes). Randomised controlled trials and prospective longitudinal studies are needed to better understand the long-term efficacy and safety issues surrounding these medicines.

Other recommendations for research

6 Using routinely collected real-world data to assess the effectiveness of continuous glucose monitoring

Based on routinely collected real-world data, what is the effectiveness and cost effectiveness of CGM devices to improve glycaemic control in adults with type 2 diabetes?

[2022]

Why the committee made the recommendation for research

The committee also made a recommendation for research on using routinely collected real-world data to assess the effectiveness and cost effectiveness of CGM. They agreed that this has the potential to show the direct effects of the technology used by people with type 2 diabetes instead of interpreting it through the results of clinical trials. Increased monitoring of routine healthcare data including registries and audits would ensure that findings from a broader population are captured.

Full details of the evidence and the committee's discussion are in

evidence review C: continuous glucose monitoring in adults with type 2 diabetes